

Lack of recovery and widespread depensation in Northwest Atlantic cod stocks

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Keywords: Atlantic Cod, stock recovery, Allee effects, population dynamics

Abstract

Limited recovery following widespread collapses of Atlantic cod (*Gadus morhua*) stocks suggests that depensatory processes may constrain rebuilding. We evaluated the presence and strength of Allee effects across 24 cod stocks using a surplus production framework that integrates recruitment, somatic growth, and natural mortality. Surplus production was derived from catch and biomass time series, and per capita production rates were analyzed as a function of biomass using segmented regressions to identify Allee effect thresholds. The strength of depensation was quantified as the relative reduction in production at low biomass compared to expected compensatory dynamics. All stocks declined below a proxy limit reference point (40% of maximum observed biomass), indicating widespread collapse; however, recovery trajectories varied substantially. Stocks that recovered generally maintained positive production at low biomass and exhibited weak or no evidence of Allee effects. In contrast, non-recovering stocks frequently displayed strong depensatory dynamics, with near-zero or negative production at their lowest observed biomass. The strength of the Allee effect was positively associated with both the severity of depletion and the likelihood of failed recovery. Some stocks without detectable Allee effects also failed to recover, reflecting persistently low productivity across their biomass range. These results demonstrate that surplus production dynamics provide a robust and integrative framework for detecting and quantifying Allee effects. Recovery potential in Atlantic cod appears to depend jointly on depletion severity, productivity at low biomass, and the strength of depensatory processes.

32 Introduction

33 Understanding the dynamics of populations following collapse is central to conservation
34 and management, as recovery potential is closely linked to per capita population growth
35 rate and can be constrained by continued overexploitation during and after collapse.
36 Under density-dependent population dynamics, per capita growth is generally expected
37 to increase as abundance declines due to reduced intraspecific competition (Nicholson
38 1933). However, below a critical threshold, this relationship may reverse, with per capita
39 growth declining as abundance decreases further. This phenomenon, known as an Allee
40 effect, introduces depensatory dynamics that can substantially slow or even prevent
41 recovery from depletion (Courchamp et al. 1999; Minto et al. 2008; Neubauer et al.
42 2013).

43 Despite their theoretical importance, Allee effects are rarely detected in empirical
44 analyses of exploited marine fish stocks (Hilborn et al. 2014). When depensation has
45 been reported, it has most often been documented in Atlantic cod (*Gadus morhua*)
46 stocks (Keith and Hutchings 2012; Winter et al. 2023), potentially reflecting the
47 widespread and prolonged depletion experienced by this species. Most empirical
48 studies have implicitly assumed that Allee effects arise through recruitment limitation at
49 low spawning biomass. Winter et al. 2023 showed that changes in water temperature
50 and/or in spawner weight of multiple cod stocks could modify the Allee effect strength,
51 defined as the relative change between maximum and minimum recruitment per capita
52 at low abundance. Depensatory dynamics can also emerge through elevated mortality
53 rates (Kuparinen et al. 2014). This distinction is particularly relevant for Northwest
54 Atlantic cod stocks, several of which have experienced substantial increases in natural
55 mortality in recent decades, including predation-driven depensation in NAFO Division
56 4TVn associated with grey seal predation (Neuenhoff et al. 2019).

57 Detecting depensatory dynamics in fisheries data using stock–recruit relationships
58 (SRRs) is widely recognized as challenging (Hilborn et al. 2014; Liermann and Hilborn
59 1997; Myers et al. 1995). For example, Myers et al. (1995) found evidence of a
60 significant Allee effect in only 3 of 128 SRRs. Similarly, an analysis of the RAM Legacy
61 database by Keith and Hutchings (2012), which examined recruits-to-SSB ratios across
62 104 species, found little evidence of Allee effects except for Atlantic cod. More recently,
63 Albertsen et al. (2025) identified Allee effects in 5 of 81 ICES stocks. Hilborn et al. 2014
64 repeated the Myers et al. 1995 analysis with 15 more years of additional data from the
65 RAM Legacy database and only found significant depensation in 4 of 113 stocks when
66 looking at the relationship between recruitment and biomass but in 8 of 109 stocks
67 when looking at the relationship between surplus production and biomass. Overall, the
68 detection of Allee effects depend on multiple factors, including species and life-history

type, time series length and stock state, data sources, and analytical approach (SRR versus surplus production).

As suggested by Walters et al. (2008), surplus production–biomass relationships provide a useful diagnostic for assessing whether stocks exhibit strong compensatory responses. Surplus production offers an integrative framework for evaluating Allee effects because it reflects the net change in biomass in the absence of fishing, incorporating the combined influences of recruitment, somatic growth, maturation, and natural mortality. Consequently, surplus production dynamics are sensitive to depensatory processes regardless of whether they arise from reduced recruitment, increased mortality, or shifts in population demography driven by fishing or environmental effects. For example, Neuenhoff et al. (2019) identified evidence of an Allee effect in NAFO 4TVn Atlantic cod using surplus production per biomass versus biomass relationships, whereas no such signal was detected using the stock–recruit relationship for the same stock. Despite these advantages, relatively few studies have used production-based approaches to detect Allee effects or to quantify the strength of depensation across multiple stocks.

Here, we use surplus production dynamics to examine the presence and strength of Allee effects across multiple Atlantic cod stocks within a common analytical framework, with particular emphasis on how productivity changes at low biomass and how these changes relate to recovery following collapse. Specifically, we assess whether cod stocks exhibit Allee effect thresholds and Allee thresholds in surplus production dynamics, quantify the strength of depensatory effects using a consistent production-based approach, and evaluate whether variation in Allee effect strength, low-biomass productivity and the level of depletion are associated with differences in historical stock recovery outcomes. We extended our investigation beyond public databases to include government stock assessment reports for cod stocks not represented in those datasets.

Methods

Atlantic Cod stocks catch and biomass data was obtained from the RAM database (Ricard et al. 2012) and stock assessment reports (Table 1). For each stock, a proxy for a biomass limit reference point (LRP) was defined as 40% of the maximum observed biomass. The stocks depletion was defined as ratio of the minimum observed biomass to the LRP. Recovered stocks was defined as stocks were biomass increased above the LRP until the terminal year of the time series, after a period of biomass lower than the LRP.

Surplus production was calculated as:

$$(1) \quad P_t = C_t + B_{t+1} - B_t$$

Where t indexes year, P is the stock surplus production, C is the fishery catch and B is the stock biomass.

Allee effects were investigated using the relationship between surplus production and biomass. Following Hutchings 2015, the Allee effect threshold was defined as the biomass where the per capita production rate (P_t/B_t) declines with declining biomass. The Allee threshold was defined as the point where the per capita production rate becomes negative as biomass decreases.

Piecewise regressions of the per capita production rate as a function of biomass were conducted using the segmented package (Fasola et al. 2018) in the R statistical software. The estimated regression breakpoint (Ψ) was used as the Allee effect threshold value for downstream analyses.

For each stocks, distinct linear regressions were fit to data for biomass below Ψ , above Ψ , and across all biomass. Stocks with estimated positive slopes before the breakpoint (depensation zone) and negative slopes after the breakpoint (compensation zone) were classified as stocks experiencing Allee effects. For stocks where a linear regression model over all biomass had lower AIC than a segmented regression, stocks where the estimated slope was positive were classified as stocks experiencing Allee effects.

For stocks classified as experiencing Allee effects with both a depensation and a compensation zone, production rates between Ψ and the lowest observed biomass (hereafter expected compensatory production rates) were predicted using the linear regression from the compensation zone. This allowed to estimate the production rates for each stock if depensation at low biomass did not occur.

For each stock, the strength of the Allee effect (As) was defined as:

$$As = \frac{pr_c - pr_d}{pr_c}$$

Where pr_c is the expected compensatory production rate at minimum observed biomass, pr_d is the predicted production rate at minimum observed biomass in the depensation zone (Figure 1). As is a relative measure of decrease in production rate from the Allee effect and can be compared across stocks.

135 *Table 1: Atlantic cod stock identification label (Stock ID), stock unit definition, year range of data and data*
 136 *source.*

Stock ID	Stock unit	Years	Source
Northwest Atlantic stocks			
2j3kl	NAFO 2J3KL	1508-2018	Schijns et al. 2021
3Pn4RS	NAFO 3Pn4RS	1973-2021	Ouellette-Plante et al. 2025
3Ps	NAFO 3Ps	1959-2020	Varkey et al. 2025
4tvn	NAFO 4TVn	1917-2023	Turcotte et al. 2024
4vsw	NAFO 4Vsw	1958-2008	Mohn and Rowe 2012
4X5Y	NAFO 4X5Y	1983-2016	Andrucschenko et al. 2022 and
5Zjm	NAFO 5Zjm	1978-2020	RAM
GB	Georges Banks	1978-2022	
EGOM	Eastern Gulf of Maine	1981-2022	
SNE	Southern New England	1981-2022	
WGOM	Western Gulf of Maine	1981-2022	
3NO	Southern Grand Banks	1953-2011	Rideout et al. 2021
1IN	NAFO Subarea 1 (inshore)	1976-2021	RAM
1F-XIV	NAFO 1F and ICES 14	1973-2021	RAM
Northeast Atlantic stocks			
BA2224	Western Baltic	1985-2021	RAM
BA2532	Eastern Baltic	1946-2021	RAM
FAPL	Faroe Plateau	1958-2022	RAM
ICE	Iceland Grounds	1955-2023	RAM
IIIaW-IV-VIId	ICES 3a(west)-4-7d	1962-2022	RAM
IS	Irish Sea	1968-2023	RAM
NEAR	North-East Arctic	1943-2021	RAM
Vla	West of Scotland	1980-2022	RAM
VIIek	Celtic Sea	1980-2021	RAM

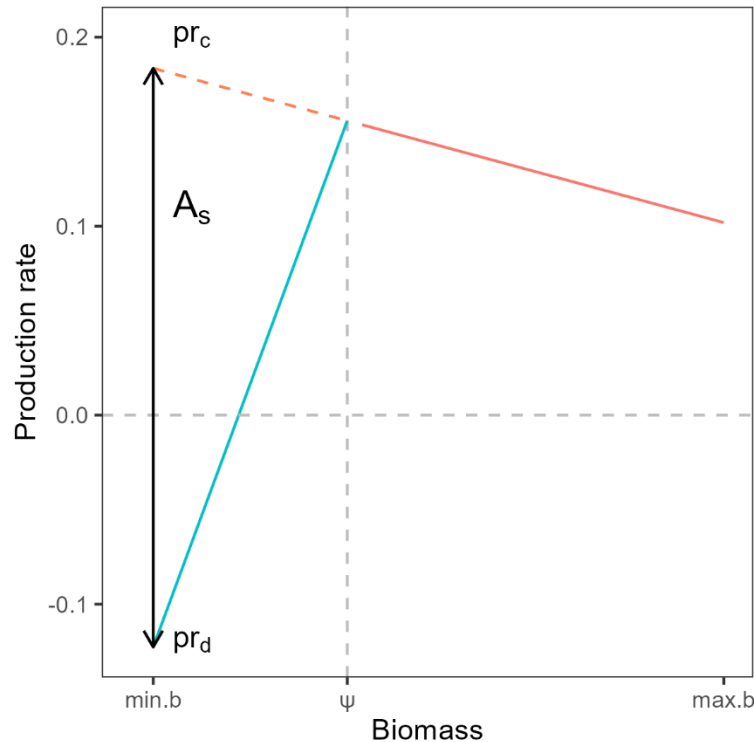


Figure 1 : Conceptual representation of estimated quantities for stocks analyses. Biomass (x axis) is the stock biomass for all years, production rate (y axis) is the per capita surplus production for all years, Ψ is the estimated breakpoint in the relationship between production rate and biomass, the red solid line is a linear regression line in the compensation zone (biomass $> \Psi$), the blue solid line is a linear regression line in the depensation zone (biomass $< \Psi$), the red dashed line is the expected compensatory production rates at biomass $< \Psi$, pr_c is the expected compensatory production rate at minimum biomass, pr_d is the production rate at minimum biomass in the depensation zone, the black vertical double arrow illustrates the distance between pr_c and pr_d , A_s is the Allee effect strength.

Results

Twenty-four Atlantic cod stocks were analyzed: 15 from the northwest Atlantic and 9 from the northeast Atlantic. For most stocks, biomass was highest early in the time series and declined over time. All stocks (100%) fell below their assumed LRP and were therefore classified as collapsed (Figure 2; Table 2). Five stocks were classified as recovered; three from the northwest Atlantic and two from the northeast Atlantic (Figure 2; Table 2). In these recovered stocks, biomass was initially high, declined below the LRP, and subsequently increased above the LRP for multiple years through to the terminal year, except for the ICE stock, where only the terminal year biomass exceeded the LRP.

Regional differences were observed in mean production and depletion levels. Mean production at low biomass (pr_d) differed significantly between regions (Wilcoxon rank-sum test: $W = 115$, $p = 0.005$), with higher values in the Northeast Atlantic. Depletion also differed between regions

(Wilcoxon rank-sum test: $W = 114$, $p = 0.006$), with Northwest Atlantic stocks generally exhibiting greater depletion.

Significant regression breakpoints (Ψ), as well as depensatory and compensatory zones, were identified for eight stocks, suggesting the occurrence of emergent Allee effects (Figure 3). For these stocks, Allee effect strength (A_s) ranged from 0.53 to 1.22. Although negative production rate observations were present in most stocks, the mean production rate at the lowest observed biomass (pr_d) was negative only for the 4TVn stock. Across these stocks, pr_d ranged from -0.06 to 0.43, and depletion ranged from 0.04 to 0.27. Of the eight stocks, only 1IN was classified as recovered.

Five stocks exhibited significant positive slopes across the full biomass range, indicating the occurrence of an Allee effect throughout the available time series (Figure 4). For these stocks, mean production at low biomass (pr_d) ranged from 0.03 to 1.22, and depletion ranged from 0.01 to 0.28. All five stocks were from the northwest Atlantic, and none were classified as recovered.

Eleven stocks exhibited either significant negative slopes or non-significant relationships and were therefore classified as not exhibiting an Allee effect (Figure 5). Across these stocks, pr_d ranged from 0.11 to 0.87, and depletion ranged from 0.04 to 0.59. Eight of these stocks were from the northeast Atlantic. Among stocks in this category, two of the three northwest Atlantic stocks and two of the eight northeast Atlantic stocks were classified as recovered.

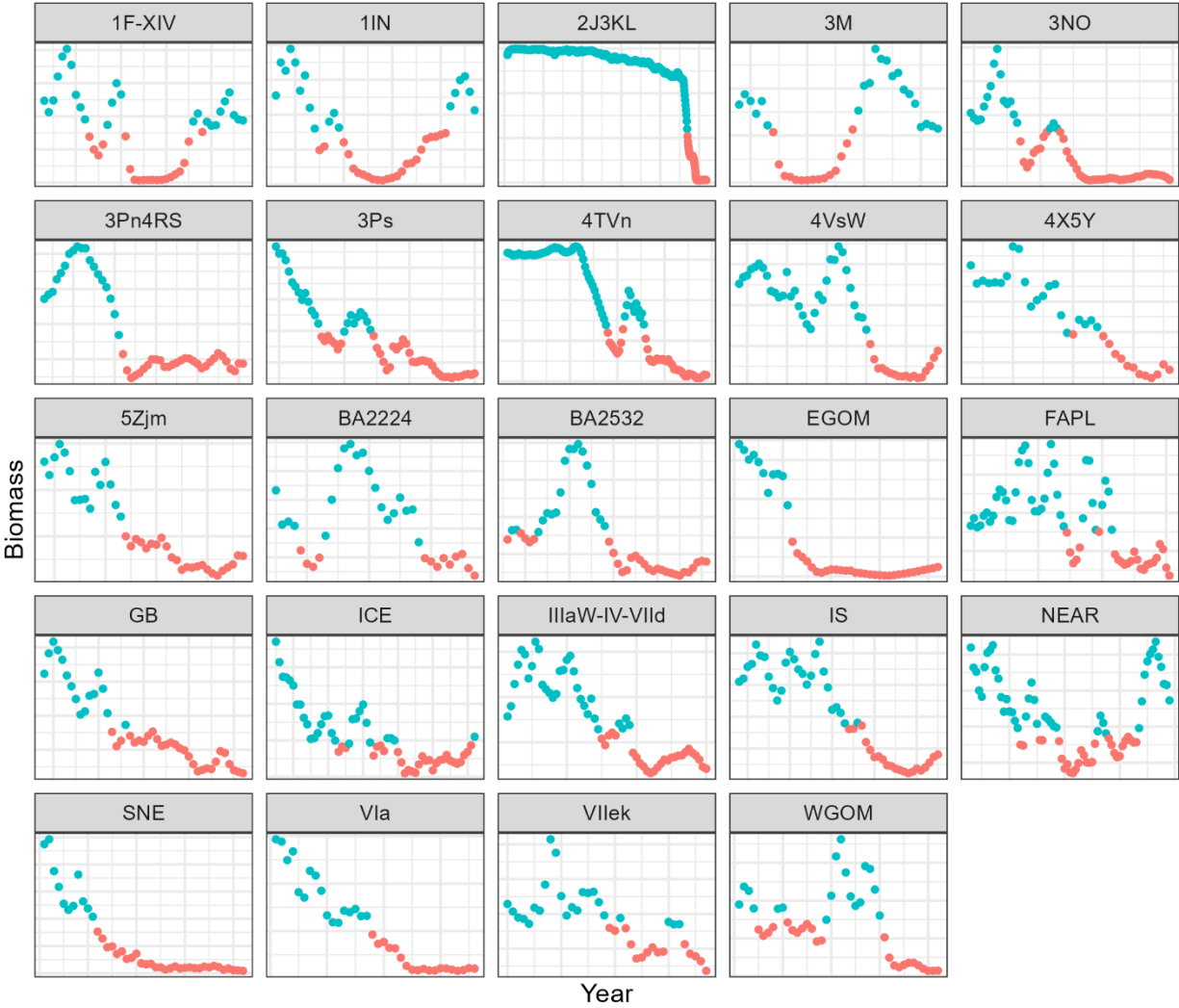
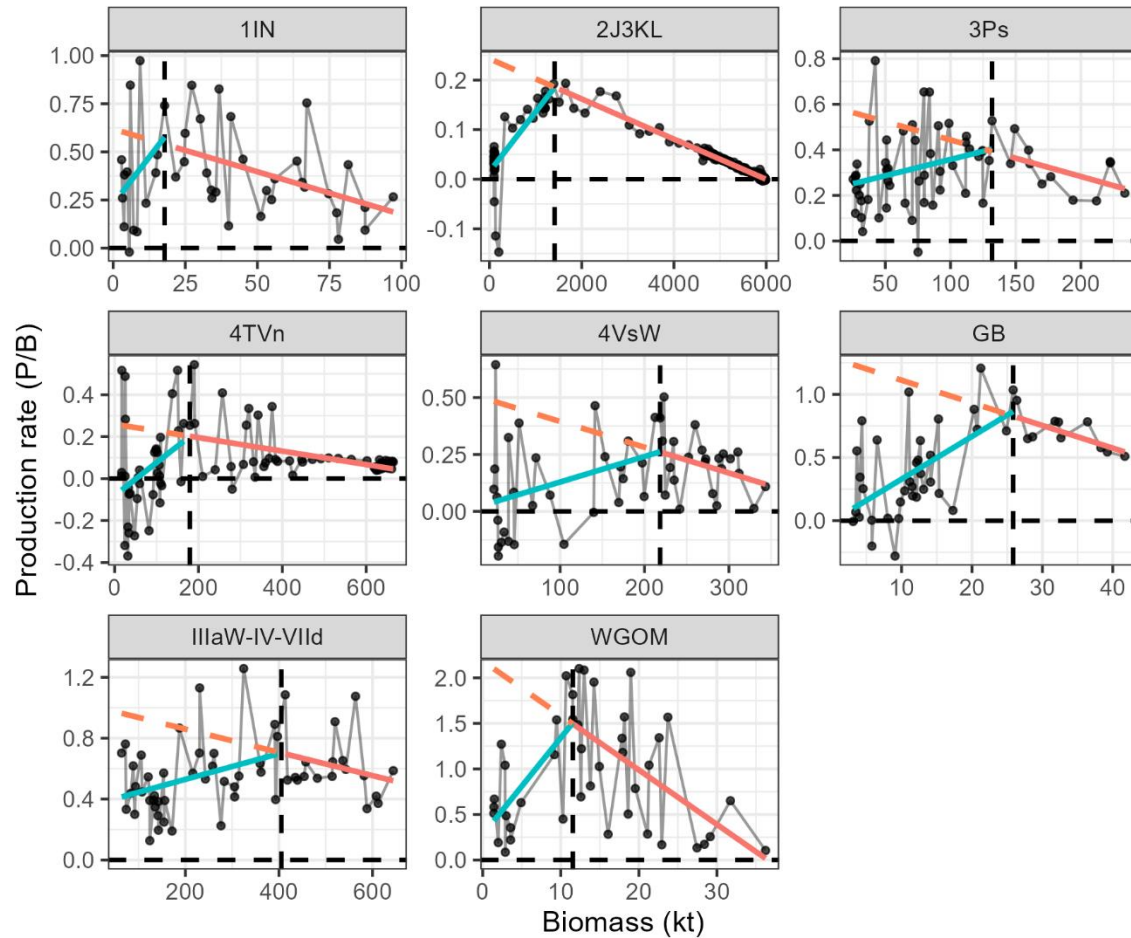


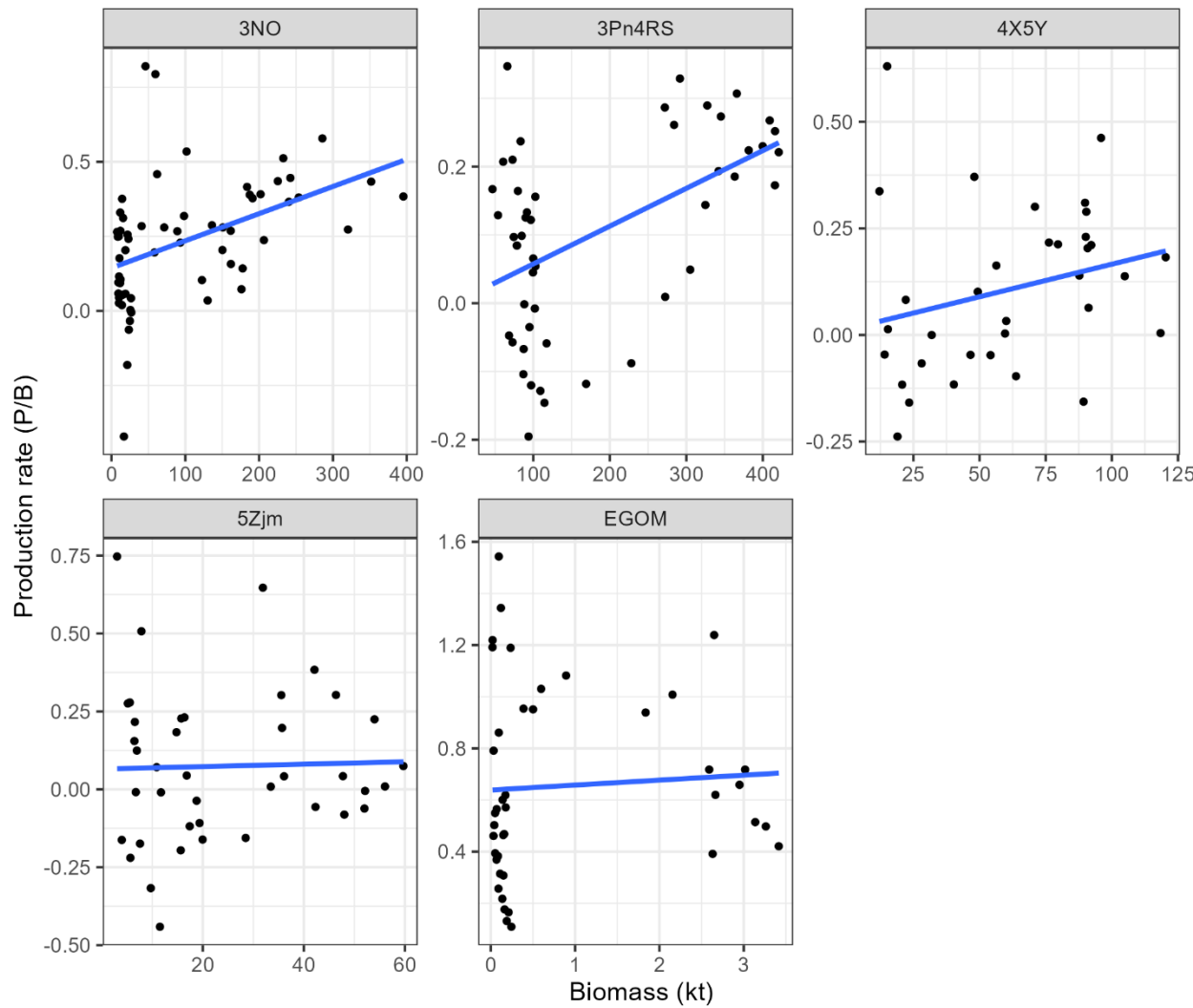
Figure 2: Time (x axis) series of biomass (y axis) for all stocks considered in the analysis, blue points are above the assumed stock biomass limit reference point and red points are below the assumed stock biomass limit reference point.



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182 Figure 3: Production rate (y axis) in function of biomass (x axis), the black vertical dashed line is the estimated
 183 breakpoint (Ψ) in the relationship between production rate and biomass, the red solid line is a linear
 184 regression line in the compensation zone (biomass > Ψ), the blue solid line is a linear regression line in the
 185 depensation zone (biomass < Ψ), the red dashed line is the expected compensatory production rates at
 186 biomass < Ψ .

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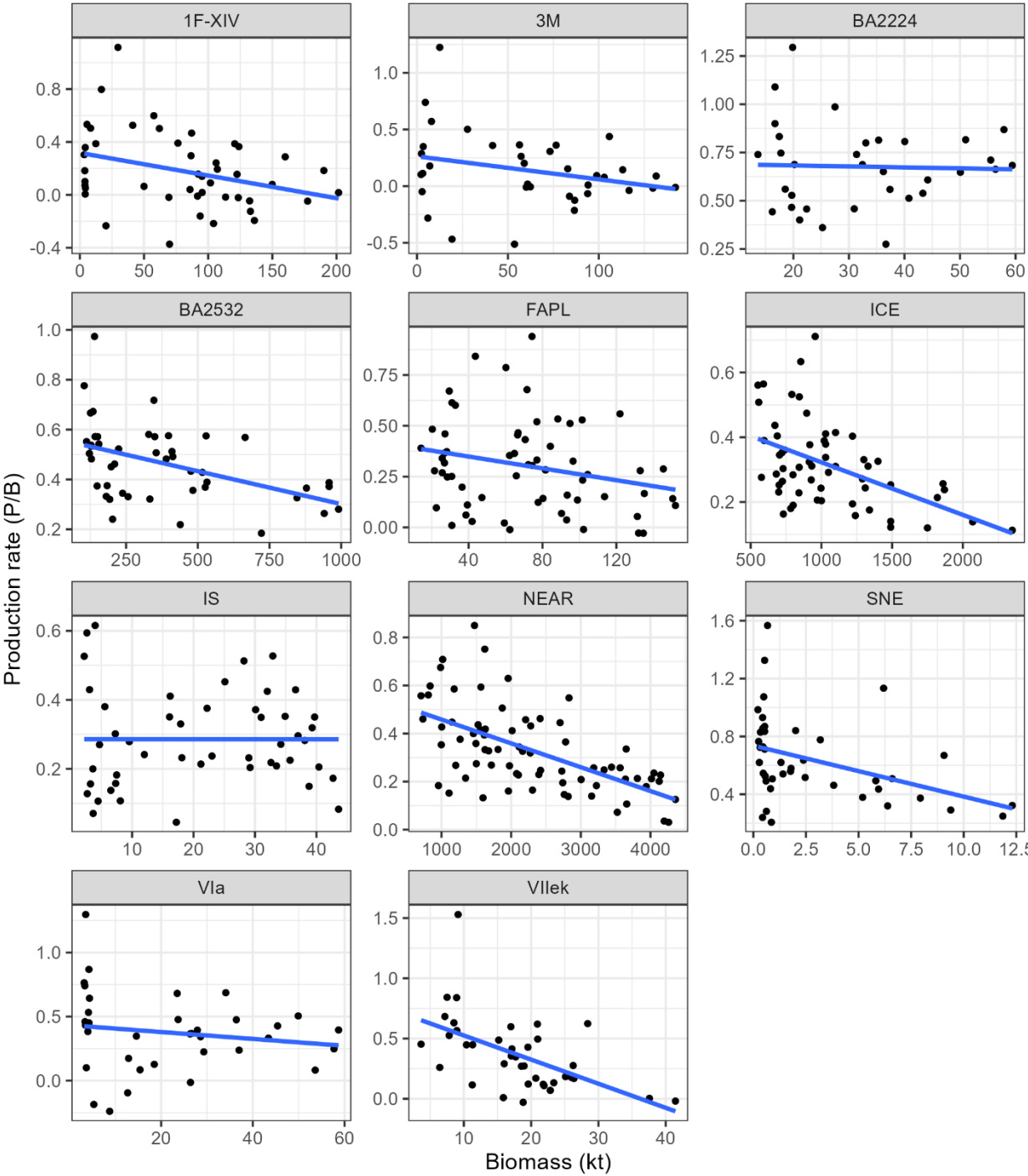
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Figure 4: Depensation stocks; Atlantic cod stocks production rate (y axis) in function of biomass (x axis) the blue solid line is a linear regression line over the range of data, showing only stock where a linear regression had a positive slope and lower AIC than a segmented regression model.



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193 Figure 5: Compensation stocks; Atlantic cod stocks production rate (y axis) in function of biomass (x axis) the

194 blue solid line is a linear regression line over the range of data, showing only stock where a linear regression

195 had a negative slope and lower AIC than a segmented regression model

Table 2: Atlantic cod stock identification label (Stock ID), Allee effect strength (A_s), production rate at lowest observed biomass, lowest depletion level (fraction of proxy for biomass limit reference point), Allee effect detected, and stock status in terminal year.

Stock ID	Allee effect	Allee effect strength (A_s)	Mean production rate at lowest observed biomass (pr_d)	Depletion	Status
Northwest Atlantic stocks					
4TVn	Yes	1.22	-0.06	0.06	Collapsed
4VSw	Yes	0.92	0.04	0.16	Collapsed
GB	Yes	0.92	0.10	0.19	Collapsed
2J3KL	Yes	0.89	0.03	0.04	Collapsed
WGOM	Yes	0.80	0.43	0.10	Collapsed
3Ps	Yes	0.55	0.25	0.27	Collapsed
1IN	Yes	0.53	0.29	0.08	Recovered
3Pn4RS	Yes	-	0.03	0.28	Collapsed
4X5Y	Yes	-	0.03	0.25	Collapsed
3NO	Yes	-	0.14	0.05	Collapsed
5Zjm	Yes	-	0.31	0.13	Collapsed
EGOM	Yes	-	1.22	0.01	Collapsed
1F-XIV	No	-	0.34	0.04	Recovered
3M	No	-	0.11	0.04	Recovered
SNE	No	-	0.80	0.04	Collapsed
Northeast Atlantic stocks					
IIIaW-IV-VIId	Yes	0.57	0.41	0.25	Collapsed
ICE	No	-	0.51	0.59	Recovered
BA2224	No	-	0.87	0.57	Collapsed
NEAR	No	-	0.51	0.41	Recovered
BA2532	No	-	0.69	0.26	Collapsed
FAPL	No	-	0.33	0.24	Collapsed
VIIek	No	-	0.73	0.22	Collapsed
VIa	No	-	0.74	0.14	Collapsed
IS	No	-	0.53	0.13	Collapsed

Discussion

Main findings

Evidence of depensation (Allee effects) was detected in 11 of the 24 Atlantic cod stocks analyzed, with a strong geographic pattern: ten occurred in the Northwest Atlantic, and only one in the Northeast Atlantic. Eight stocks exhibited depensation below a breakpoint in the production–biomass relationship, while five showed consistent depensation across the full biomass range. By using surplus production dynamics, this study provides a quantitative and comparable measure of Allee effect strength across stocks. Stocks that recovered generally maintained positive production at low biomass and showed weak or no depensation, indicating that recovery potential is jointly influenced by depletion severity, low-biomass production, and the strength of depensatory dynamics.

Regional differences

Northwest Atlantic stocks were characterized by biomass depletion to lower levels, lower production at low biomass and high occurrence of depensation when compared to northeast Atlantic stocks. This spatial contrast is consistent with previous findings indicating that reduced recovery potential has largely been confined to Northwest Atlantic cod stocks, whereas stocks in the Eastern Atlantic, particularly those around mainland Europe, generally exhibit higher recovery potential (Keith et al. 2026). These regional differences have been attributed to factors such as fisheries-induced evolution, shifts in life-history traits (e.g., age and size at maturity), the loss of large, old spawners, and environmental conditions in the Western Atlantic (see Keith et al. 2026 and references therein). These findings are consistent with recent evidence of Allee effects in specific Northwest Atlantic cod stocks (e.g., Perälä et al. 2022), further supporting the prevalence of depensatory dynamics in this region

Depletion and Recovery

All stocks reached very low relative biomass levels, where their minimum biomass relative to a proxy for a LRP ranged between 1% and 59% across stocks. Under these levels of depletion, all stocks were classified as collapsed at one point over time. Despite broadly similar depletion trajectories, recovery outcomes varied substantially among stocks.

Stocks that recovered generally maintained positive production at their lowest observed biomass and exhibited weak or no evidence of depensation. For example, ICE and NEAR showed compensatory dynamics across their observed biomass range and retained positive production at low biomass, enabling recovery from collapse. Recovery was also observed for 1F-XIV and 1IN at much lower depletion levels (< 0.1). 1F-XIV did not show depensation and maintained compensatory dynamics, allowing recovery.

The 1IN stock displayed the weakest depensation of all stocks and maintained positive production at low biomass, allowing recovery.

In contrast, many stocks that failed to recover exhibited strong depensation, indicated by positive A_s values. These stocks also exhibited very low or negative production at their lowest observed biomass. Stocks such as 3Pn4RS, 4X5Y, 4vsw, 4TVn and 2J3KL showed strong depensation, with production values near zero or negative (≤ 0.04). These stocks were also among the most severely depleted (depletion < 0.3), suggesting that compensatory dynamics constrained population growth following collapse to low levels, as suggested by Hutchings 2015.

Stocks indicating no detectable Allee effect, showed mixed recovery outcomes. Several such stocks (e.g., BA2224, BA2532, Vllek, Vla, 5Zjm, IS, and 3NO) did not recover despite the absence of depensation, likely reflecting low production across their biomass range, including at their lowest observed biomass, and thus weak compensatory dynamics. It is possible that the time series for these stocks are from highly depleted states, masking a potential breakpoint in per capita production rates. In contrast, 1F-XIV recovered, emphasizing the importance of positive production at low biomass even when depensation is weak or absent.

Overall, recovery potential appears to be jointly influenced by depletion severity, production at low biomass, and the strength of compensatory dynamics. Stocks experiencing extreme depletion combined with strong depensation and near-zero or negative production at low biomass consistently failed to recover. Conversely, recovery was most likely when production remained positive at low biomass, even when weak depensation was present.

Production analyses and depensation strength

A key contribution of this study is the use of surplus production dynamics to quantify the strength of depensation. Expressing depensation in production units provides a more intuitive and comparable metric than estimating additional parameters in stock recruitment relationships, which rely on assumptions that may not hold in practice. In particular, stock recruitment relationships assume that recruitment is primarily driven by spawner abundance, yet many stocks exhibit weak or inconsistent recruitment dependence once environmental variability is considered.

Here, depensation was identified either through a breakpoint in the production–biomass relationship or as a positive slope across the observed biomass range. Stocks exhibiting a positive slope over their full range may reflect truncated time series lacking observations at higher biomass, where a breakpoint might otherwise be detectable. We detected depensation in two stocks where Allee effects have been previously identified

either using stock recruit relationship (3PS Cod; Perälä et al 2022) or a production-biomass relationship (4TVn Cod ; Neuenhoff et al. 2019)

More generally, limited data at both low and high stock sizes often constrain the ability to estimate stock recruitment relationships curvature or steepness, and non-stationary dynamics driven by environmental variability or regime shifts can further undermine inference. Under these conditions, fitting stock recruitment relationships can introduce structural uncertainty and potentially obscure underlying population processes.

Previous large-scale analyses based on stock recruitment relationships have generally found limited evidence for depensation (e.g., Myers et al. 1995; Liermann and Hilborn 1997; Hilborn et al. 2014), likely reflecting both low statistical power at low abundance and the focus on recruitment as the sole pathway for density dependence (Hutchings and Rangeley 2011). By contrast, the surplus production approach used here captures depensatory processes acting across multiple vital rates, including growth, maturity and survival, and may therefore provide greater sensitivity for detecting emergent depensation.

Although biomass estimates are derived from stock assessment models and thus inherit model assumptions, the use of per capita surplus production provides a relatively direct and semi-empirical characterization of population dynamics. In contrast to fully parametric surplus production models (e.g., Schaefer-type), this approach does not impose a fixed functional form on the production–biomass relationship, reducing the risk that model structure masks depensatory dynamics.

As in previous studies, the ability to detect depensation remains constrained by the availability of observations at very low biomass, and stocks that collapsed prior to the assessment era may be underrepresented.

Implications for stock assessment and management

Breakpoints in the production–biomass relationship (Ψ), interpreted here as thresholds for depensatory dynamics, provide a biologically grounded indicator of impaired stock productivity. For example, such a breakpoint has been used to identify serious harm in 4TVn Atlantic cod (Turcotte et al. 2026), consistent with theoretical expectations that depensation emerges at low abundance and can constrain recovery (Courchamp et al. 1999; Hutchings 2014, 2015). These thresholds therefore offer a practical basis for defining biologically meaningful limit reference points.

In Canada, the LRP is intended to operationalize the objective of avoiding serious harm to fish stocks (DFO 2009). Stocks at or below their LRPs are considered to be at unacceptable risk of impaired productivity or reproductive capacity, including reduced recruitment, growth, or survival, and diminished resilience to perturbations (Shelton and Rice 2002;

311 DFO 2023). Within this framework, the breakpoints estimated here can be interpreted as
312 empirical indicators of the onset of serious harm.

313
314 Incorporating these thresholds into reference point setting could improve the alignment
315 between assessment outputs and management objectives by explicitly accounting for
316 compensatory dynamics (Hutchings 2014). In particular, LRPs set above estimated
317 breakpoints would reduce the risk of entering states where productivity is impaired and
318 recovery potential is limited, thereby supporting more precautionary and biologically
319 informed management (Hutchings 2015).

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References

- Albertsen, C. M., Perälä, T., Cardinale, M., Winker, H., & Trijoulet, V. (2025). Rebuilding and Reference Points Under Compensatory and Depensatory Recruitment: A Meta-Analysis of Northeast Atlantic Fish Stocks. *Fish and Fisheries*, 26(1), 171–192. <https://doi.org/10.1111/faf.12872>
- Andrushchenko, I., Clark, D., Irvine, F., MacEachern, E., Martin, R. and Wang, Y. 2022. 2018 4X5Y Atlantic Cod Framework Data Inputs. DFO Can. Sci. Advis. Sec. Res. Doc. 2022/006.v + 86 p.
- Courchamp, F., Clutton-Brock, T., and Grenfell, B. 1999. Inverse density dependence and the Allee effect. *Trends Ecol. Evol.* 14: 405–410
- DFO. 2009. A fishery decision-making framework incorporating the precautionary approach. Last updated 2009-03-23.
- DFO. 2019. Stock Assessment of Atlantic Cod (*Gadus morhua*) in NAFO Divisions 4X5Y. DFO Can. Sci. Advis. Sec. Sci. Advis. Rep. 2019/015.
- DFO. 2023. Science advice on guidance for limit reference points under the fish stocks provisions. DFO Can. Sci. Advis. Sec. Sci. Advis. Rep. 2023/009.
- Fasola S, Muggeo VM, Kuchenhoff H (2018). “A heuristic, iterative algorithm for change-point detection in abrupt change models.” *Computational Statistics*, 33(2), 997-1015.
- Hilborn, R., Hively, D.J., Jensen, O.P., and Branch, T.A. 2014. The dynamics of fish populations at low abundance and prospects for rebuilding and recovery, *ICES Journal of Marine Science*; 71,8, 2141-2151, <https://doi.org/10.1093/icesjms/fsu035>
- Hutchings, J. A., & Rangeley, R. W. (2011). Correlates of recovery for Canadian Atlantic cod (*Gadus morhua*). *Canadian Journal of Zoology*, 89(5), 386–400. <https://doi.org/10.1139/Z11-022>
- Hutchings, J. A. (2014). Renaissance of a caveat: Allee effects in marine fish. *ICES Journal of Marine Science*, 71(8), 2152–2157. <https://doi.org/10.1093/icesjms/fst179>
- Hutchings, J.A. 2015. Thresholds for impaired species recovery. *Proc. R. Soc. B Biol. Sci.* 282(1809). doi:10.1098/rspb.2015.0654.
- Keith, D.M., and Hutchings, J.A. 2012. Population dynamics of marine fishes at low abundance. *Can. J. Fish. Aquat. Sci.* 69(7): 1150–1163. doi:10.1139/F2012-055.
- Keith DM, Bowlby HD, Albouy C, Auber A, Baribeau DM, Boyce DG, et al. (2026) Demographics and recovery potential of exploited marine teleosts. *PLoS One* 21(1): e0340369. <https://doi.org/10.1371/journal.pone.0340369>
- Kuparinen, A., Keith, D. M., & Hutchings, J. A. (2014). Allee effect and the uncertainty of population recovery. *Conservation Biology*, 28(3), 790–798. <https://doi.org/10.1111/cobi.12216>
- Liermann, M., and Hilborn, R. 1997. Depensation in fish stocks: a hierarchic Bayesian meta-analysis. *Can. J. Fish. Aquat. Sci.* 54(9):1976–1984. doi:10.1139/cjfas-54-9-1976.
- Minto, C., Myers, R. A., and Blanchard, W. 2008. Survival variability and population density in fish populations. *Nature*, 452: 344–347
- Mohn, R.K., and Rowe, S. 2012. Recovery Potential Assessment for the Laurentian South Designatable Unit of Atlantic Cod (*Gadus morhua*): The Eastern Scotian Shelf Cod Stock (NAFO Div. 4VSW). DFO Can. Sci. Advis. Sec. Res. Doc. 2011/138: viii + 71p.

- 362 Myers, R.A., Barrowman, N.J., Hutchings, J.A., and Rosenberg, A.A. 1995. Population
363 dynamics of exploited fish stocks at low population levels. *Science*, 269(5227): 1106–1108.
364 doi:10.1126/science.269.5227.1106.
- 365 Nicholson, A. 1933. The balance of animal populations. *J. Anim. Ecol.* 2: 131–178
- 366 Neubauer, P., Jensen, O. P., Hutchings, J. A., and Baum, J. K. 2013. Resilience and recovery of
367 overexploited marine populations. *Science*, 340: 347–349.
- 368 Neuenhoff, R.D., Swain, D.P., Cox, S.P., Mcallister, M.K., Trites, A.W., Walters, C.J.,
369 and Hammill, M.O. 2019. Continued decline of a collapsed population of Atlantic Cod. *Can.*
370 *J. Fish. Aquat. Sci.* 76: 168–184
- 371 Ouellette-Plante, J., Benoît, H.P., and Lussier, J.-F. 2025. The Status of the Northern Gulf of St.
372 Lawrence (3Pn, 4RS) Atlantic Cod (*Gadus morhua*) Stock in 2022. DFO Can. Sci. Advis. Sec.
373 Res. Doc. 2025/039. xx + 244 p.
- 374 Perälä, T, Hutchings, J.A., and Kuparinen, A. 2022. Allee effects and the Allee-effect zone in
375 northwest Atlantic cod. *Biol. Lett.* 18(2): 20210439. doi:10.1098/rsbl.2021.0439
- 376 Ricard, D., Minto, C., Jensen, O.P. and Baum, J.K. (2012) Evaluating the knowledge base and
377 status of commercially exploited marine species with the RAM Legacy Stock Assessment
378 Database. *Fish and Fisheries* 13 (4) 380-398. DOI: 10.1111/j.1467-2979.2011.00435.x
- 379 Schijns, R., Froese, R., Hutchings, J.A., and Pauly, D. 2021. Five centuries of cod catches in
380 Eastern Canada. *ICES J. Mar. Sci.* 78(8): 2675–2683. Oxford University Press.
381 doi:10.1093/icesjms/fsab153.
- 382 Shelton, P.A., and Rice, J.C. 2002. Limits to overfishing: reference points in the context of the
383 Canadian perspective on the precautionary approach. DFO Can. Sci. Advis. Sec. Res. Doc.
384 2002/084. 29p.
- 385 Turcotte, F., McDermid, J.L., and Ricard, D. 2024. Scientific requirement for the rebuilding plan
386 of southern Gulf of St. Lawrence (NAFO Division 4TVn) Atlantic Cod (*Gadus morhua*). DFO
387 Can. Sci. Advis. Sec. Res. Doc. 2024/053(July): iv + xx.
- 388 Turcotte, F., McDermid, J. L., Ricard, D., & Swain, D. P. (2026). Defining a biomass limit
389 reference point under time-varying productivity: a case study of Atlantic cod in the southern Gulf
390 of St. Lawrence, Canada. *Canadian Journal of Fisheries and Aquatic Sciences*, 83, 1–12.
391 <https://doi.org/10.1139/cjfas-2025-0147>
- 392 Varkey, D., Wheeland, L.J., Regular, P., Rideout, R., Kumar, R., Rogers, B., and Vigneau, J.
393 2025. Analytical Assessment for the Status of Atlantic Cod (*Gadus morhua*) Stock in NAFO
394 Subdivision 3Ps in 2021. DFO Can. Sci. Advis. Sec. Res. Doc. 2025/043. iv + 48 p.
- 395 Walters, C. J., Hilborn, R., & Christensen, V. 2008. Surplus production dynamics in declining
396 and recovering fish populations. *Canadian Journal of Fisheries and Aquatic Sciences*, 65(11),
397 2536–2551. <https://doi.org/10.1139/F08-170>
- 398 Winter, A.M., Vasilyeva, N., and Vladimirov, A. 2023. Spawner weight and ocean temperature
399 drive Allee effect dynamics in Atlantic cod, *Gadus morhua*: inherent and emergent density
400 regulation. *Biogeosciences* 20(17): 3683–3716. doi:10.5194/bg-20-3683-2023.